CipherVault CLI — Official User Documentation

Secure File Encryption System (Standalone EXE Version)

---

1. Overview

CipherVault CLI is a standalone, offline file-encryption tool designed to protect sensitive data using modern cryptographic standards.
The application is packaged as a single **Windows executable (.exe)** and does not require installation, internet access, or Node.js.

CipherVault uses:

- **AES-256-GCM** encryption
- **PBKDF2** secure key derivation
- **Authenticated encryption** with integrity validation
- **Secure deletion** of original plaintext files

The system is interactive and allows users to browse folders, select files, and perform encryption or decryption through a controlled menu interface.

---

2. Running CipherVault (EXE Version)

To start the application:

1. Locate **CipherVault.exe**
2. Double-click the file
3. The application launches in a terminal window and displays the interactive menu

No installation is required.
No command line knowledge is required.
The program executes directly from any folder on your system.

---

3. Interactive Mode (Main Interface)

Upon startup, CipherVault displays:

- Application banner
- A three-option menu:

```
[1] Encrypt a file
[2] Decrypt a file
[3] Exit
```

Selecting encryption or decryption opens a **folder navigator** that allows the user to browse:

- Desktop
- Documents
- Downloads
- Pictures
- Videos
- Music
- Any custom folder path

The navigator automatically shows:

- Only raw files when encrypting
- Only *.enc* files when decrypting

This prevents user mistakes and ensures correct operation.

---

4. How to Encrypt a File

1. Open **CipherVault.exe**
2. Select **Encrypt a file**
3. Navigate to the folder containing your file
4. Select the file
5. Enter and confirm your password
6. CipherVault encrypts the file and outputs:

```
filename.ext → filename.enc
```

What happens internally:

- A secure password-derived 256-bit key is generated
- File is encrypted using AES-256-GCM streaming
- Authentication tag is stored to prevent tampering
- A metadata header is written for safe restoration
- The original plaintext file is irreversibly overwritten and deleted (secure-delete)

---

5. How to Decrypt a File

1. Open **CipherVault.exe**
2. Select **Decrypt a file**
3. Navigate to a folder containing encrypted files (*.enc*)
4. Select a file
5. Enter the password used during encryption

CipherVault restores the file:

```
filename.enc → filename.ext
```

If the password is incorrect or the file is altered, decryption will fail to protect file integrity.

---

6. Password Policy

CipherVault enforces a strict password standard:

- Minimum length: **10 characters**
- Must include:
  - Letters
  - Numbers
- Password and confirmation must match
- Weak passwords are rejected

Passwords are **never stored** and cannot be recovered.
Losing the password means permanent loss of access to encrypted data.

---

7. Secure-Delete Mechanism

After encryption completes:

1. The plaintext file is overwritten with zeros
2. The file is overwritten again with random bytes
3. The file is physically deleted

This prevents:

- File recovery
- Forensic reconstruction
- Undelete operations

This ensures the original data no longer exists unencrypted anywhere on disk.

---

8. Integrity & Security Features

CipherVault includes:

- Encrypted metadata structure
- Authentication tag validation (AES-GCM)
- Salted password-based key derivation
- Automatic restoration of file extensions
- Strict file-type filtering to prevent errors
- Offline-only operation for privacy
- Full local execution — no external dependencies

---

9. Notes for Users

- CipherVault runs on Windows systems only (EXE build).
- The tool works entirely offline; no network traffic is generated.
- The program cannot recover a forgotten password.
- Encrypted files are portable and can be decrypted on any machine running CipherVault.